

Project Documentation: Automated Schema Evolution with AWS Glue and Athena

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Problem Statement

A data engineering team is responsible for managing and analyzing a continuously growing and evolving dataset of **customer transactions** stored in **Amazon S3**. Since the dataset schema may change over time (e.g., adding new columns), manual updates are time-consuming and error-prone. The team uses **Amazon Athena** for querying this data, so they need a solution that can automatically detect and apply schema changes to maintain query accuracy.

Objective

- Automate schema management for evolving data in Amazon S3.
 - Integrate with Amazon Athena to enable seamless querying of updated data.
 - Ensure zero manual intervention for schema updates.
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Architecture & Tools Used

Component	Purpose
Amazon S3	Stores raw transaction data in CSV format
AWS Glue Crawler	Scans data in S3 and updates schema in Data Catalog
AWS Glue Data Catalog	Stores metadata and schema for Athena queries
Amazon Athena	Queries data using updated schema from Glue Data Catalog

Step-by-Step Implementation

Step 1: Prepare Sample Data with Schema Changes

Create and upload the following two sample files to your S3 bucket at: `s3://your-bucket/customer-transactions/`

Sample Files:

- transactions_v1.csv
- transactions_v2.csv

transactions_v1.csv:

```
transaction_id,customer_id,amount
1,101,200.50
2,102,150.00
3,103,400.75
```

transactions_v2.csv:

```
transaction_id,customer_id,amount,transaction_type
4,104,300.00,online
5,105,250.00,offline
6,106,175.25,online
```

Step 2: Set Up AWS Glue Crawler

1. Navigate to **AWS Glue > Crawlers > Add Crawler**
2. **Name:** customer-transactions-crawler
3. **Source:** Choose data source as **S3**, point to s3://your-bucket/customer-transactions/
4. **IAM Role:** Choose or create a role with Glue & S3 permissions
5. **Output:** Create or choose a **Glue Database**, e.g., customer_db
6. **Schedule:** Optional – you can run on-demand or automate via EventBridge
7. Run the crawler

 The crawler will scan all files and automatically detect schema changes, storing the updated schema in the Glue Data Catalog.

Step 3: Query Using Amazon Athena

1. Go to **Amazon Athena** console
2. Select database: customer_db
3. Run a query:

```
SELECT * FROM customer_transactions LIMIT 10;
```

Athena uses the **Glue Catalog table**, so it dynamically reflects the schema updates from the latest crawler run.

Step 4 (Optional): Automate Schema Detection

- Set up an **EventBridge rule** to trigger the Glue Crawler when new files are uploaded to S3.
- You can also use **Lambda** to trigger the crawler programmatically.

Benefits

-  **Automated schema detection** – No need to manually define schema changes
-  **Seamless querying** – Athena uses the latest schema via Glue
-  **Cost-efficient** – No need to spin up servers
-  **Production-ready** – Easily scales to large datasets

Real-World Use Cases

- Financial transactions (banking)
- E-commerce order data
- Customer behavior analytics with evolving formats

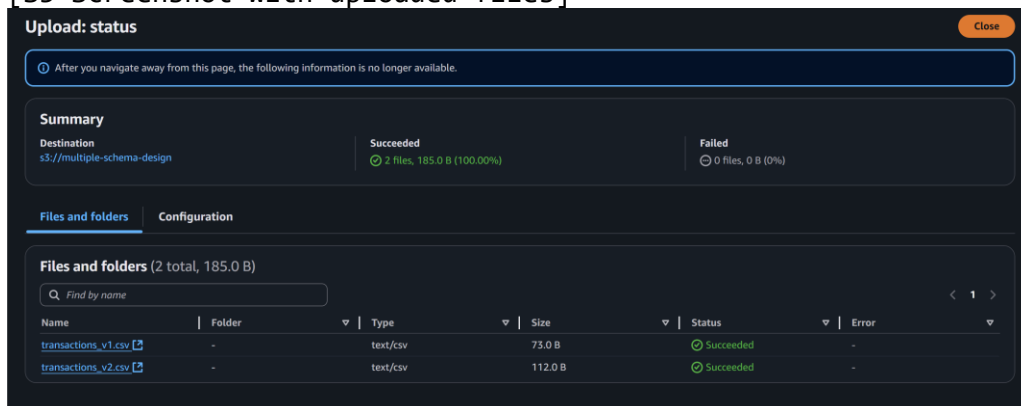
Project Folder Structure

```
s3://your-bucket/  
└─ customer-transactions/  
    └─ transactions_v1.csv  
    └─ transactions_v2.csv
```

Workspace Output (Paste Your Results Below)

 Paste screenshots, query outputs, S3 structure, job logs, etc. here:

[S3 Screenshot with uploaded files]



The screenshot shows the 'Upload: status' window in the AWS Glue console. It displays a summary of the upload process, indicating that 2 files (185.0 B) were successfully uploaded to the S3 bucket 's3://multiple-schema-design'. Below the summary, the 'Files and folders' tab is selected, showing a table with the following data:

Name	Folder	Type	Size	Status	Error
transactions_v1.csv	-	text/csv	73.0 B	Succeeded	-
transactions_v2.csv	-	text/csv	112.0 B	Succeeded	-

[Glue crawler run screenshot or log snippet]

The screenshot displays the AWS Glue console interface. On the left, the navigation pane shows the 'Data Catalog' and 'Data Integration and ETL' sections. The main panel shows the 'my-multiple-schema' crawler details. A green banner at the top indicates 'Crawler successfully starting'. The crawler properties are as follows:

Property	Value
Name	my-multiple-schema
IAM role	my-data-combine-role
Database	multiple_schema
State	READY
Description	-
Security configuration	-
Lake Formation configuration	-
Table prefix	-
Maximum table threshold	-

Below the properties, the 'Crawler runs' tab is selected, showing a single run that completed successfully on June 22, 2025, at 04:48:31. The run details show a duration of 52 seconds and 1 table change.

[Athena query outputs after schema detection]

The screenshot shows the Amazon Athena console. The left sidebar displays the 'Data' section with the 'multiple_schema_design' table selected. The main panel shows the SQL query editor with the following query:

```
1 SELECT *
2 FROM multiple_schema_design
3 LIMIT 10;
```

The query has been executed successfully, and the results are displayed in a table with 6 rows. The first row is highlighted.

#	transaction_id	customer_id	amount	transaction_type
1	1	101	200.5	

✓ Conclusion

By using **AWS Glue Data Catalog** in combination with **Glue Crawlers** and **Amazon Athena**, we created a robust, automated solution for managing schema changes in an evolving S3 dataset. This design ensures minimal maintenance and real-time data querying capabilities, making it ideal for scalable data engineering pipelines.



Tags

#AWS #GlueCrawler #Athena #SchemaEvolution #AutomatedETL #DataEngineering